

# Yujie Liang

Technical Designer

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## Career Objective

Seeking a Technical Designer / Gameplay Technical Designer role.

Experienced with Unreal Engine 5 Blueprint, UMG gameplay UI, C++ technical projects, and team-based game development.

Able to break down gameplay requirements into system logic, state flow, and player feedback, then prototype, debug, and support implementation directly in engine.

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## Project Experience

### Third-person Air Combat Gameplay Feedback & HUD Prototype

Technical Designer / Gameplay UI Designer  
Mar 2026 – May 2026

Implemented an air combat gameplay feedback prototype using UE5 Blueprint and UMG, including radar, crosshair, lock-on indicators, damage warning, and tutorial UI. Connected enemy detection, attack range, player health, and UI states to help players identify targets, assess threats, and understand combat status during high-speed aerial gameplay.

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### CPU Rasterizer Performance Optimisation Project

C++ / Visual Studio / AVX2 / Multithreading  
Oct 2025 – Dec 2025

Implemented multiple real-time rendering optimisation techniques in a CPU rasterizer, including lighting optimisation, backface culling, vertex transform caching, edge-function rasterisation, SIMD, and multithreaded tile rendering. Used Visual Studio Profiler to analyse performance bottlenecks and built a basic understanding of performance budgets and feature costs in real-time projects.

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## Contact

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## About Me

MSc Games Engineering student with a focus on gameplay systems and in-engine prototyping. Good at breaking gameplay rules into variables, states, events, and UI feedback, then quickly validating them through Blueprint. Able to communicate and bridge implementation needs between design, programming, UI, and art.

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## Education

University of Warwick  
MSc Games Engineering  
2025 – 2026

### Relevant areas:

Unreal Engine 5  
Game systems design  
C++ game development  
Real-time rendering and optimisation  
Team-based game development

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## **Path Tracing Renderer**

C++ / Rendering / BVH / BSDF

Mar 2026 – Apr 2026

Implemented path tracing modules including direct lighting, shadow testing, BVH acceleration, material sampling, and basic tone mapping. Improved C++ code reading, system breakdown, debugging, and technical communication skills through this project.

## **Skills**

C++

Unreal Engine 5

Git Collaboration

Gameplay UI